

Intro & motivation

fd - fin dim
ss - semisimple
hw - highest weight

Lie theory: Repr of fd ss complex Lie alg

Verma modules (universal hw modules)



Category $\mathcal{O}$ (BGG) - "built around" Verma modules

Prototype of hw theory

$\leadsto$ hw categories

- $\mathcal{O}$ describes e.g. repr of $SL_n(\mathbb{C})$ viewed as IR Lie group s.t. the cpt subgroup $SU_n(\mathbb{C})$ acts semisimply with finite multiplication (Harish-Chandra theory)
- Nice homological properties
- Solves for example character formulas for irreducible hw modules.

- Connected to
- Coxeter groups, Kazhdan-Lusztig theory
 - geometry of flag varieties G/P ^{reductive} $\leftarrow$ parabolic
 - new example: diagrammatical categories
 - Categorification: many examples come from category $\mathcal{O}$ (eg refined knot invariants)
 - Prototype or motivation for many constructions in current repr theory
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Course plan

Part I (Stroppel):

- basic categories

- Def & properties of category $\mathcal{O}$
 - HW categories (Cline-Paishall-Scott, Ringel 1980's)
 - Modern formulation: Brundan-Stroppel (2022)
- $\leadsto$ 6 functor formalism

Part II (J. Gruber):

- Repr theory of alg groups
- Kac-Moody Lie alg from hw POV

1. Categorical background

Def $\mathcal{A}$ -category is abelian if

- Additive
- preadditive (0-obj, $\text{Hom}_{\mathcal{A}}(A, B)$ -ab group, composition is $\mathbb{Z}$ -linear)
 - finite product
 - (co)kernels exist
 - $\text{coker}(\ker) \cong \ker(\text{coker})$

Def $\mathcal{A}$ ab category

$K_0(\mathcal{A}) = \text{Grothendieck group of } \mathcal{A} :=$

free abelian group generated by isoclasses

$[M]$ of $M \in \mathcal{A}$ modulo $[C] = [A] + [B]$

if $\exists \text{ SES } A \hookrightarrow C \twoheadrightarrow B$

Rank: $\mathcal{A}$ additive then $K_0^{\oplus}(\mathcal{A}) := \text{split G group}$

= free ab group $\iff \exists \text{ split SES } A \hookrightarrow C \twoheadrightarrow B$

ex: The full subcategory of projectives in $\mathcal{A}$ (which ab) is additive.

- Similar definition of K_0 for any exact category in Quillen space
 - even for Waldhausen cats $A \hookrightarrow C \twoheadrightarrow B$ cofibrated seq.
- just a side note.

Lemma 1.1 $\mathcal{A}, \mathcal{B}$ abelian (additive)

$f: \mathcal{A} \rightarrow \mathcal{B}$ exact (additive) then f induces

group homomorphism $\bar{f}: K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B})$ ($K_0^\oplus(\mathcal{A}) \rightarrow K_0^\oplus(\mathcal{B})$)

with $\bar{f}([M]) = [f(M)]$

Ex: 1) $\mathcal{A} = \text{Vec}_k$ then $K_0(\mathcal{A}) \cong \mathbb{Z}$

$$[k] \longleftarrow 1$$

$$[V] \longmapsto \dim V$$

$$[k^n] \longleftarrow n \in \mathbb{N}$$

formal inverse in $K_0(\mathcal{A}) \Rightarrow -[k^n] \longleftarrow -n$

2) $\mathcal{A} = \text{Rep}(\text{sl}_2(\mathbb{C})) = \text{fd repr of } \text{sl}_2(\mathbb{C})$

$K_0(\mathcal{A}) \cong \mathbb{Z}^{\mathbb{N}} := \text{free } \mathbb{Z}\text{-module w/ basis } b_i, i \in \mathbb{N}_0$

$[L(n)] \mapsto b_n$ irred module of $\dim n$ $n \in \mathbb{N}_0$

Rank: In these cases $K_0(\mathcal{A}) \cong K_0^\oplus(\mathcal{A})$

because $\mathcal{A}$ is SS.

Lemma 1.2 $\mathcal{A}$ ab of finite length.

Let $\{L_i\}_i$ be a system of repr of isoclasses of irreducibles

Then $K_0(\mathcal{A}) \cong \mathbb{Z}^I := \text{free } \mathbb{Z}\text{-module with } b_i$

$$[L_i] \mapsto b_i$$

$$[M] \mapsto \sum \alpha_i b_i \quad \text{almost all } \alpha_i = 0$$

with $\alpha_i = [M : L_i] = \text{multiplicities of } L_i \text{ in a JH-series of } M$.

Goal: Thm 1.3 $\mathcal{A}$ ab of finite length

Then $\exists$ bijection

$$\left\{ \begin{array}{l} \text{indecomposable} \\ \text{projective} \\ \text{objects} \end{array} \right\} \xrightarrow{[-1]} \left\{ \begin{array}{l} \text{simple} \\ \text{objects} \end{array} \right\} \quad \text{i.e. isoclasses of simple objects}$$

$\cong$ $\cong$

———— first half of the lecture ————

Ex $R = \mathbb{C}[x]/(x^2)$ $\mathcal{A} = R\text{-mod} = \text{fd } R\text{-modules}$

Let $\mathbb{C}$ be the irreducible 1-dim quotient module

(i.e. x acts by zero) then we have SES

$$\begin{array}{ccc} \mathbb{C} & \hookrightarrow & R \twoheadrightarrow \mathbb{C} \\ & \downarrow & \downarrow \\ & 1 & \mapsto 1 \\ & 1 & \mapsto x \end{array} \quad \text{in } \mathcal{A} \text{ (not split)}$$

Every irred object in $\mathcal{A}$ is isomorphic to $\mathbb{C}$

Thus $K_0(\mathcal{A}) \cong \mathbb{Z}$

Note: $[R] = [\mathbb{C}] + [\mathbb{C}] = 2[\mathbb{C}]$

$$[\mathbb{C}] \mapsto 1$$

$$[M] \mapsto \dim M$$

In particular $\mathcal{A} \cong \mathcal{B} \Rightarrow K_0(\mathcal{A}) \cong K_0(\mathcal{B})$

$\nLeftarrow$ in general

Consider $R = \mathbb{C}[X]/(X^2)$ as graded ring

$\deg(X) = 1$, $\mathcal{A} := R\text{-g mod} = \text{fd graded } R\text{-modules}$

Let $\mathbb{C}$ be 1-dim irred quotient of regular(graded) R -modules R

So $\mathbb{C}$ is concentrated in degree 0 (w/ X acting by 0)

The irreducible objects in $\mathcal{A}$ are of the form

$\mathbb{C}\langle i \rangle$ $i \in \mathbb{Z}$ (1-dim R -module concentrated in degree i
 X acts by 0)

$K_0(\mathcal{A}) \cong \mathbb{Z}[q, q^{-1}]$ viewed as free ab group w/ basis q^i $i \in \mathbb{Z}$

$$[\mathbb{C}\langle i \rangle] \longleftarrow q^i$$

$$[M] \longmapsto \sum_{h \in \mathbb{Z}} \dim M_h q^h$$

Note There is an action of $\mathbb{Z}$ on $\mathcal{A}$ by
grading shifting, $i \in \mathbb{Z}$

$$\langle i \rangle : \mathcal{A} \rightarrow \mathcal{A}$$

$$M \longmapsto M\langle i \rangle \quad ((M\langle i \rangle)_n := M_{n-i})$$

identity on morphism

$\langle i \rangle$ functors are exact and by Lemma 1.1

they induce a morphism of groups $K_0(\mathcal{A}) \rightarrow K_0(\mathcal{A})$

$$\langle i \rangle : K_0(\mathcal{A}) \rightarrow K_0(\mathcal{A})$$

$$\begin{array}{ccc} \text{||s} & \cdot q^i & \text{||s} \\ \mathbb{Z}[q, q^{-1}] & \longrightarrow & \mathbb{Z}[q, q^{-1}] \end{array}$$

The $\mathbb{Z}$ -action induces a $\mathbb{Z}[q, q^{-1}]$ action on $K_0(\mathcal{A})$
 s.t. $K_0(\mathcal{A}) \cong \mathbb{Z}[q, q^{-1}]$ is an iso of $\mathbb{Z}[q, q^{-1}]$ -modules

Next: Detour on projective covers

Def $\mathcal{A}$ abelian, $P \in \mathcal{A}$, $X \in \mathcal{A}$

$P \xrightarrow{\varphi} X$ projective cover of X if

- P projective, • φ -epi, • φ -essential

i.e. $P \xrightarrow{\varphi} X$ s.t. $\varphi \circ \alpha$ epi then α is epi
 if $\begin{array}{c} \alpha \uparrow \\ Q \end{array}$ (intuition: P is sort of the "smallest")

Lemma 1.4 proj - $P \xrightarrow{\varphi} X$ in ab $\mathcal{A}$. Then φ is a projective cover

($\Rightarrow$) $\forall \alpha \in \text{End}_{\mathcal{A}}(P)$ with $\begin{array}{c} P \xrightarrow{\varphi} X \\ \alpha \uparrow \nearrow \varphi \\ P \end{array}$ $\varphi \circ \alpha$ epi
 α iso

Proof: " $\Rightarrow$ " $\varphi \circ \alpha = \varphi \xRightarrow{\varphi \text{ epi}} \alpha$ epi $\begin{array}{ccc} P & \xrightarrow{\alpha} & P \\ \nwarrow & & \uparrow \text{id} \\ \exists \alpha' & & \hookrightarrow \text{by proj} \end{array}$
 in particular α' is mono

Now $\varphi = \varphi \circ \alpha \circ \alpha' = \varphi \circ \alpha' \xRightarrow{\varphi \text{ epi}} \alpha'$ epi

$\Rightarrow \alpha'$ iso $\Rightarrow \alpha$ iso

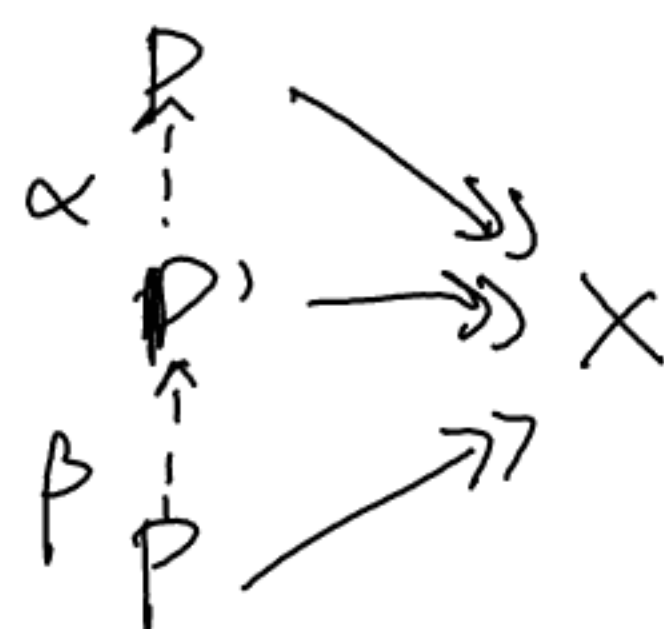
" $\Leftarrow$ " $\begin{array}{ccc} P & \xrightarrow{\varphi} & X \\ \alpha \uparrow & & \nearrow \varphi \\ Q & \xrightarrow{\varphi \circ \alpha} & X \end{array}$ $\varphi \circ \alpha$ epi $\exists \alpha' \xrightarrow{\text{by proj}} P \xrightarrow{\alpha} X$

$\alpha \circ \alpha'$ iso so α epi $\Rightarrow P \xrightarrow{\varphi} X$ proj cover QED

Cor 1.5 If proj covers exist then $\exists!$ up to iso

Proof:
$$\begin{array}{ccc} P & \xrightarrow{\varphi} & X \\ & \searrow & \uparrow \\ & & P \xrightarrow{\varphi'} X \end{array}$$
 proj covers

By proj



α, β - epi
but $\alpha \circ \beta$ is an iso by lemma 1.4
 $\Rightarrow \beta$ mono and thus iso

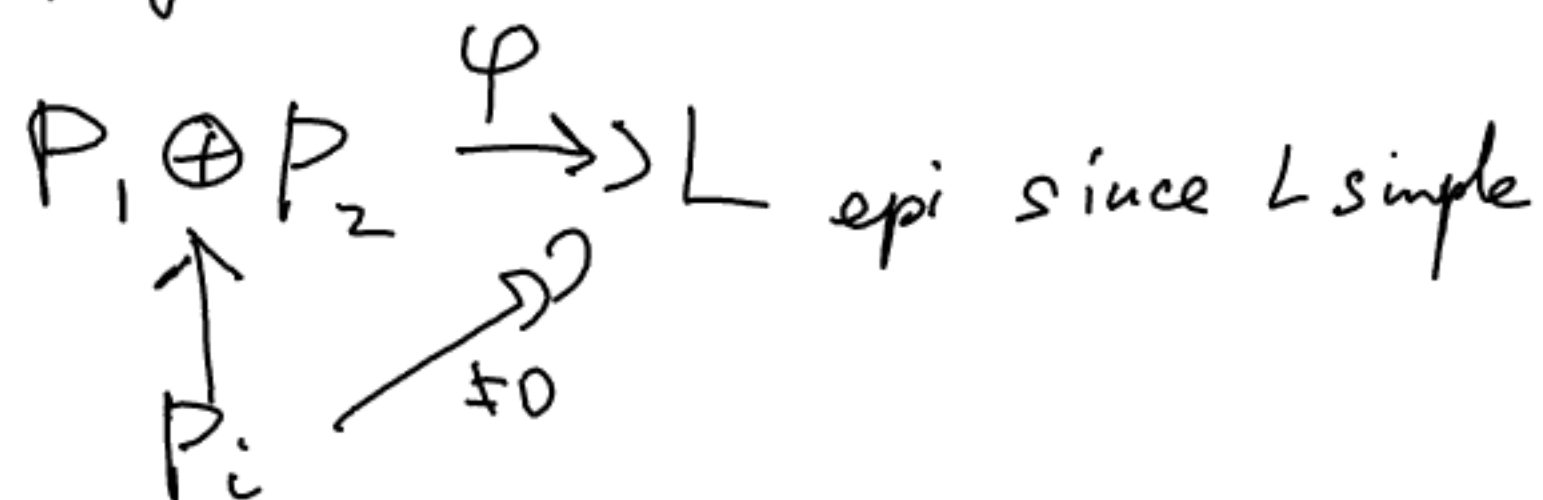


Existence?

- Projectives covered by $\text{id} : P \rightarrow P$
- R is called semiperfect if every irred R -module has a proj cover
- Exercise: A fd algebra $A = A\text{-mod}$ then every simple object has a proj cover (and then in fact every object)
- A ab $P \xrightarrow{\varphi} L$ proj cover of L simple then P is indecomposable.

Assume $P = P_1 \oplus P_2$ for proper subobjects P_1, P_2

P_i are projective



qess

$\Rightarrow \sim$

is epi $\Rightarrow P_i = P$

